

Git & GitHub Complete Guide for Students

From Your First Commit to a
Professional **GitHub Portfolio**



Git



GitHub



VS Code



GitHub Desktop



ThinkHub

Daksh Khandelwal

Founder, ThinkHub

THINK. LEARN. GROW.



Code



Commit



Push



Achieve



Git & GitHub Complete Guide for Students

From Your First Commit to a Professional GitHub Portfolio

Git

GitHub

VS Code

GitHub Desktop



Git ≠

- ▶ Version Control System
- ▶ Runs locally on your machine
- ▶ Tracks changes & creates commits
- ▶ Manages branches
- ▶ Can work entirely offline



**Git = The Engine
(Version Control)**



GitHub

- ▶ Online Git Platform
- ▶ Primarily cloud-based
- ▶ Hosts repositories & portfolios
- ▶ Enables Pull Requests, Issues & Code Review

**GitHub = The Platform
(Hosting & Collaboration)**



The Git Mental Model

EDIT → STAGE → COMMIT → PUSH



Set Up Git on Your Computer



Terminal - bash

Check Installation

```
$ git --version  
git version 2.39.2.windows.1
```

Configure Your Professional Identity

```
$ git config --global user.name "Your Name"  
$ git config --global user.email "you@example.com"
```

Verify

```
$ git config --global user.name  
Your Name
```



Do NOT put passwords or API keys in your Git config.
This identity is permanently attached to your commits.

From Project Folder → Git Repository

Before:

my-ai-project/
├── app.py
├── model.py
└── README.md

After:

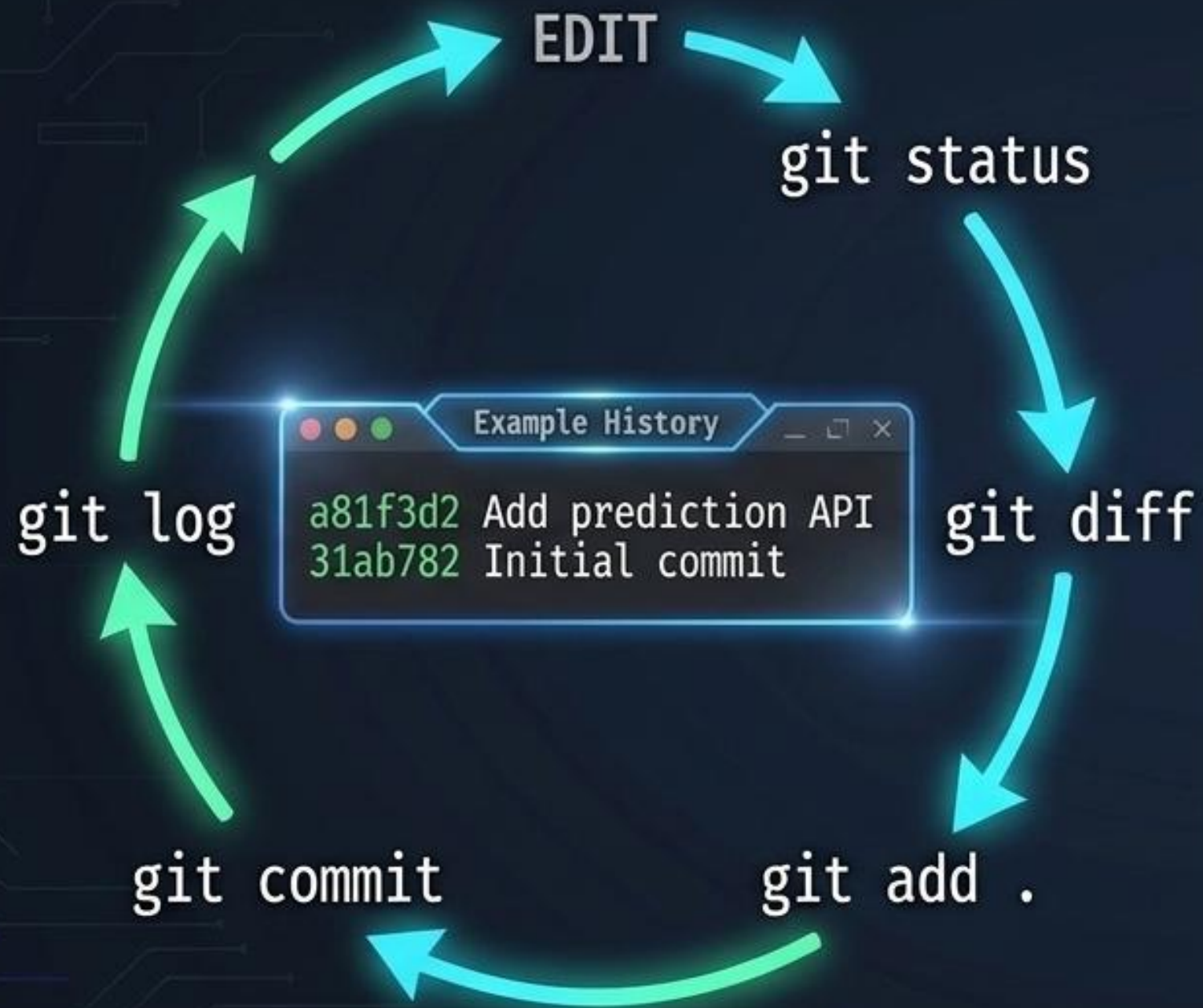
my-ai-project/
├── .git/
├── app.py
├── model.py
└── README.md

```
Terminal  
$ cd my-ai-project  
$ git init  
$ git status
```

Next Steps

```
$ git add .  
$ git commit -m "Initial commit"
```


Track Your Changes Like a Professional



Good Commits



Add authentication

Fix API validation

Communicates exactly what changed.

Bad Commits



update changes

final

asdf

Useless to your team or future self.

Connect Your Local Project to GitHub



Terminal

Step 1: Create an empty repo on GitHub (e.g., ai-resume-analyzer)

Step 2 (Connect): `$ git remote add origin https://github.com/USERNAME/REPO.git`

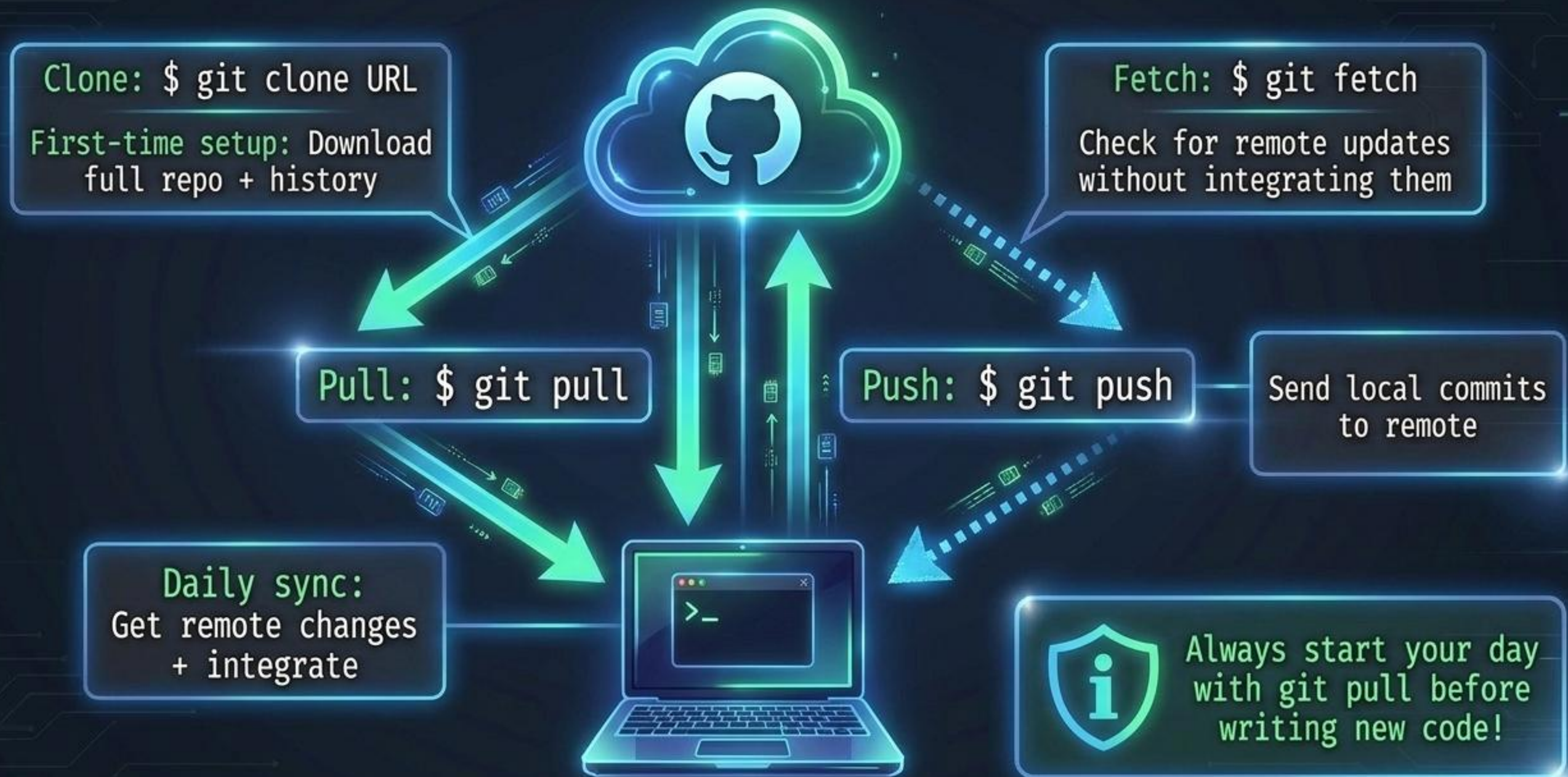
Step 3 (Verify): `$ git remote -v`

Step 4 (Branch Setup): `$ git branch -M main`

Step 5 (Push): `$ git push -u origin main`

origin is simply the standard name for your remote repository on GitHub.

Working With a Remote Repository



Branches & Merging: Work Without Breaking Main



```
Subway Map
```

```
Create & Switch: $ git switch -c feature-login
Return to Main:  $ git switch main
Merge Feature:   $ git merge feature-login
Clean Up:        $ git branch -d feature-login
```

`main` = Stable, production-ready code.
`feature` = Safe experimentation and development.

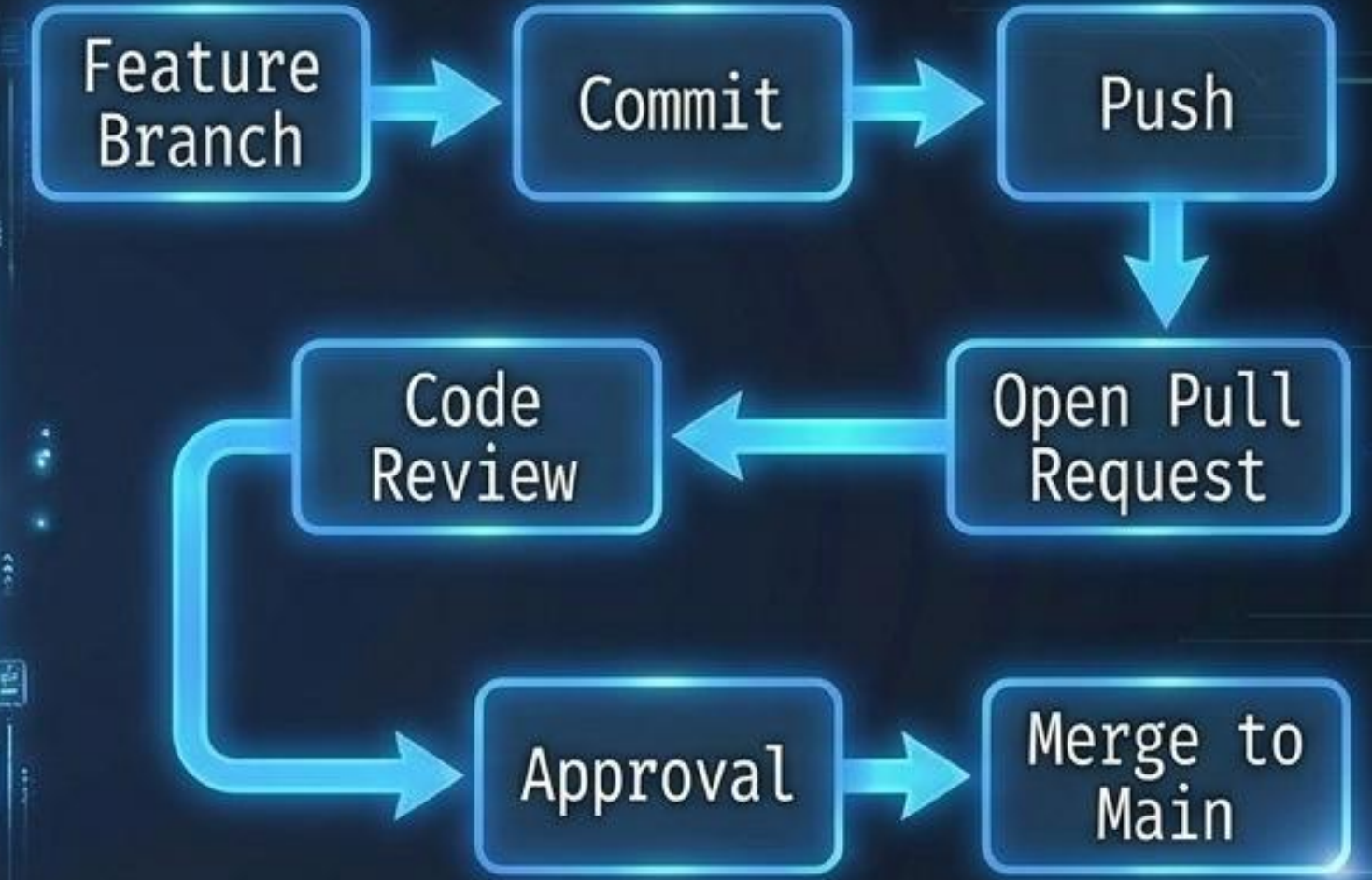
Collaboration: Merge Conflicts & Pull Requests



```
<<<<<<< HEAD
name = "Daksh"
=====
name = "Khushi"
>>>>>>> feature
```



Git is just asking you to pick a version.
Edit the file, save, and git commit.



Example PR: feature-dashboard → main
| Title: Add analytics dashboard.

GitHub Desktop: Visual Git Workflow

Current Repository &
Current Branch
(Easily switch contexts)

Changes
(Visual diffs of
modified code)

Commit to main box
(Write message and
commit)



Push origin button
(Sync with GitHub)

Git without memorizing every command. It uses the exact same Git engine under the hood, just replacing typing with clicking.

The Developer Workflow Inside VS Code

 VS Code GUI

 Terminal Equivalent



You never actually have to leave VS Code.
VS Code GUI ↔ Git Terminal ↔ GitHub

One-Click: VS Code → GitHub

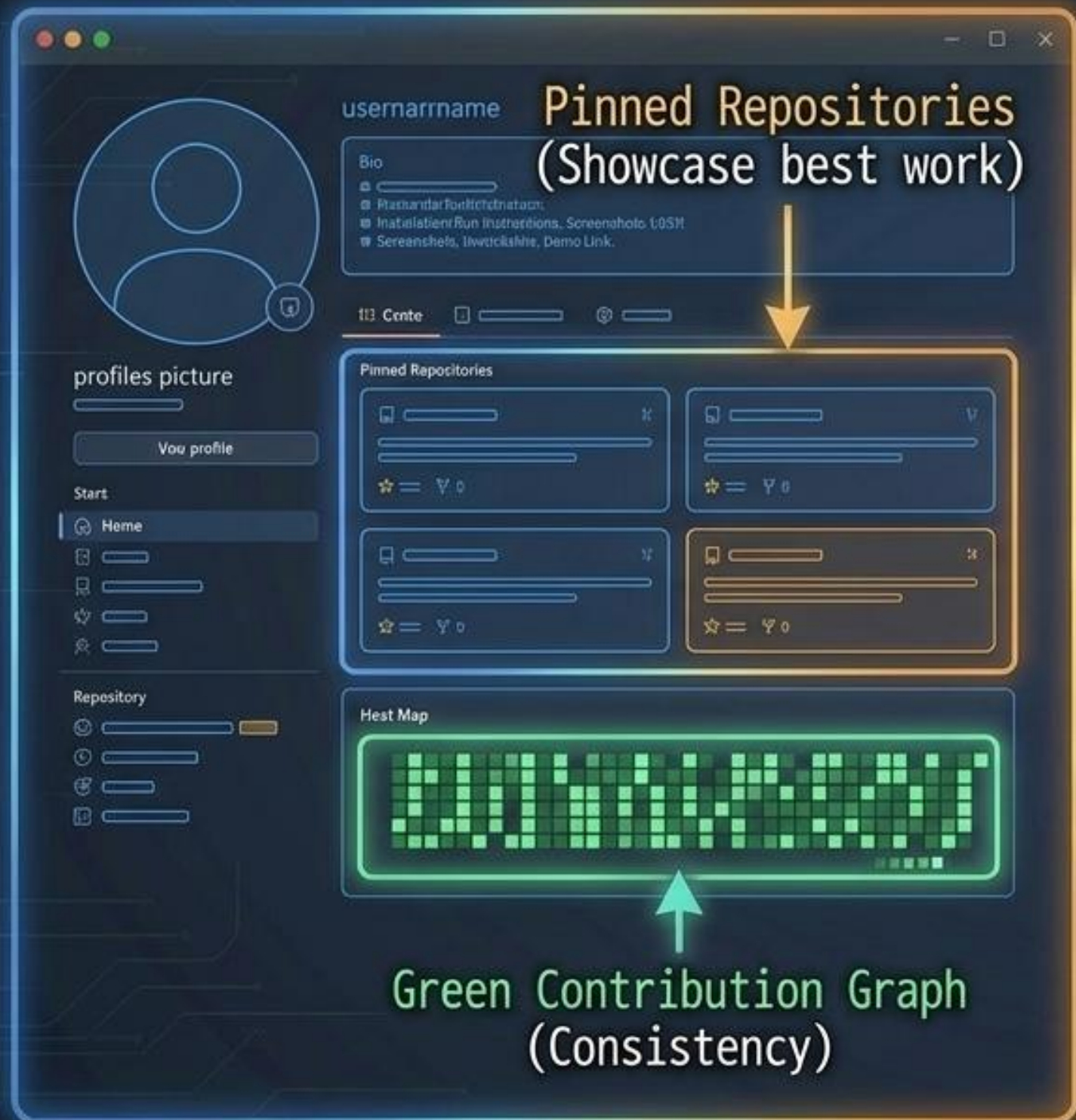


What VS Code just did for you behind the scenes:

```
git init + git add . + git commit + git branch -M main + git remote add + git push -u
```

Multiple ways to accomplish the same workflow (Terminal, VS Code, Desktop). The underlying Git concepts remain exactly the same.

Build a Professional GitHub Portfolio



The Professional README Must-Haves

- 📖 Project Overview, Features, Tech
- 🔧 Stack, Installation/Run Instructions,
- 🖼️ Screenshots, Demo Link.

Do's & Don'ts Table

DO ✓

Meaningful commits
Clean README
Use .gitignore

DON'T ✗

Expose API keys (.env)
Upload node_modules
Treat GitHub as an unfinished dumping ground

Your Git Journey Starts Here

The Cheat Sheet

<code>\$ init / clone URL</code>	Setup
<code>\$ status / diff</code>	Check
<code>\$ add . / commit -m</code>	Save
<code>\$ branch / switch -c</code>	Branching
<code>\$ pull / push / merge</code>	Syncing

The Workflows

Solo Core Workflow



Professional Team Workflow



Daksh Khandelwal | Founder, ThinkHub | Think. Learn. Grow.

Take one project today. Initialize Git. Write a README. Push it to GitHub.



THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



PRESENTED BY

DAKSH KHANDELWAL

FOUNDER, THINKHUB



LIKE

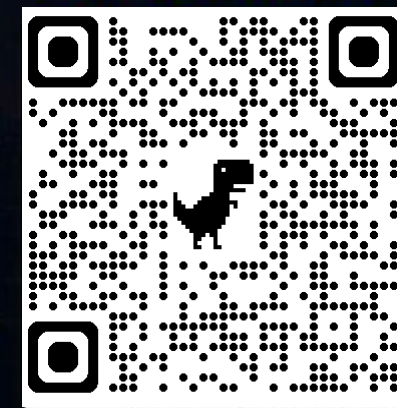


SUBSCRIBE



COMMENT

CONNECT WITH ME



Scan to connect on
LinkedIn

KEEP LEARNING. KEEP BUILDING. KEEP GROWING.